

Newton Baskaran

Design portfolio

Mechanical design, product development,
manufacturing, engineering analysis

I turn ideas into mechanical parts and working prototypes,
connecting the design with how it gets made.

EXPLORE THE PROJECTS

- | | |
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| 01 Mining truck commonality | 02 Bharath Formula Karting |
| 03 Foundry component design | 04 Portable ultrasonic nebulizer |
| 05 Plate-fin heat sink design | 06 NACA 4412 CFD analysis |
| 07 Vacuum bottle thermal reliability | 08 CFD mesh convergence |
| 09 Wind farm feasibility & design | 10 CSP heat-transfer fluid research |
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Background and Contact

FEATURED DESIGN WORK



01 / MINING TRUCK COMMONALITY



02 / BHARATH FORMULA KARTING

01 / MECHANICAL DESIGN

Mining truck commonality study

Caterpillar · Manufacturing Engineer, contract through Midsys Inc. · October 2024–October 2026



One of the mining trucks I worked on.

The design challenge

Find opportunities to reuse and standardize parts across the large mining truck lineup, beginning without prior documentation.

MY OWNERSHIP

- Initiated and led the commonality study from scratch without existing documentation.
- Identified parts that could be used on other trucks in the same lineup.
- Designed parts where there were opportunities for reuse and standardization.
- Worked with design teams across vehicle systems to develop commonality opportunities.

Project focus

Part reuse and targeted design changes to support standardization across machines, with cost-reduction opportunities identified through the study.

02 / VEHICLE DESIGN & FABRICATION

Bharath Formula Karting go-kart

September 2020–January 2021 · Team build and business leadership



Competition photograph.

155cc

Engine displacement

20

Engineering team members

The build

Collaboratively engineered and manufactured a go-kart for the National Formula Karting event, working within competition requirements.

My contribution

- Participated in mechanical design, performance calculations, and fabrication with the 20-member engineering team.
- Led the business management team in developing and presenting an industrial business plan.

COMPETITION OUTCOME

Best Business Proposal Award

03 / FOUNDRY DESIGN & PROCESS STANDARDIZATION

Foundry component design & standardization

Caterpillar · Chills, cores, and basins · Design, reverse engineering, and supplier-part procurement



Mapleton Foundry

Design to procurement

Created a process flow for supplier-part procurement in collaboration with quality and manufacturing engineers.

The work

Designed foundry components and standardized supplier-part procurement, working with quality and manufacturing engineers to support production needs.

MY CONTRIBUTION

- Designed chills, cores, and basins to support foundry casting operations.
- Reverse engineered selected components according to the use case.
- Created models and drawings with specified tolerances to support manufacturability.
- Helped with the PPAP process in collaboration with quality and manufacturing engineers.
- Created and standardized a process flow for supplier-part procurement.

04 / PRODUCT DEVELOPMENT

Portable ultrasonic nebulizer

Arizona State University · Engineering Intern · June–October 2024

The project

Developed a portable ultrasonic nebulizer from the ground up, translating a product concept into working prototypes and planning for scalable manufacturing.

6

Engineers on the team

2024

Development at ASU

Why this work matters

Creating a compact nebulizer was a cause I cared about. The goal was to make the device easier to carry and use, with the potential to help more people living with asthma.

MY OWNERSHIP

- Led a multidisciplinary team across design, prototyping, and manufacturing phases.
- Led product layout development and working prototypes.
- Worked with outsourcing companies to source essential components.
- Created and managed the bill of materials and considered health and safety requirements.

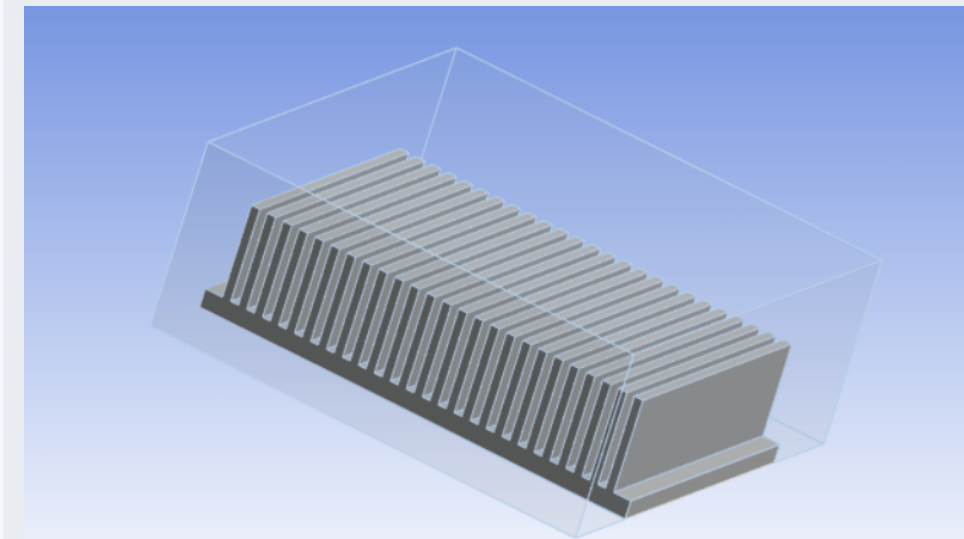
DEVELOPMENT SCOPE

Concept & layouts
Working prototypes
Supplier coordination & BOM

05 / THERMAL DESIGN & PREDICTIVE MODELING

Parallel plate-fin heat sinks

ANSYS simulation + MATLAB machine learning · September–November 2023



Heat sink geometry used in the simulation study.

10

Simulation setups

100-2000

Reynolds range

R > 0.9

ANN correlation

Engineering question

How do fin geometry and flow conditions affect heat dissipation and pressure drop—and can a trained model help predict those responses?

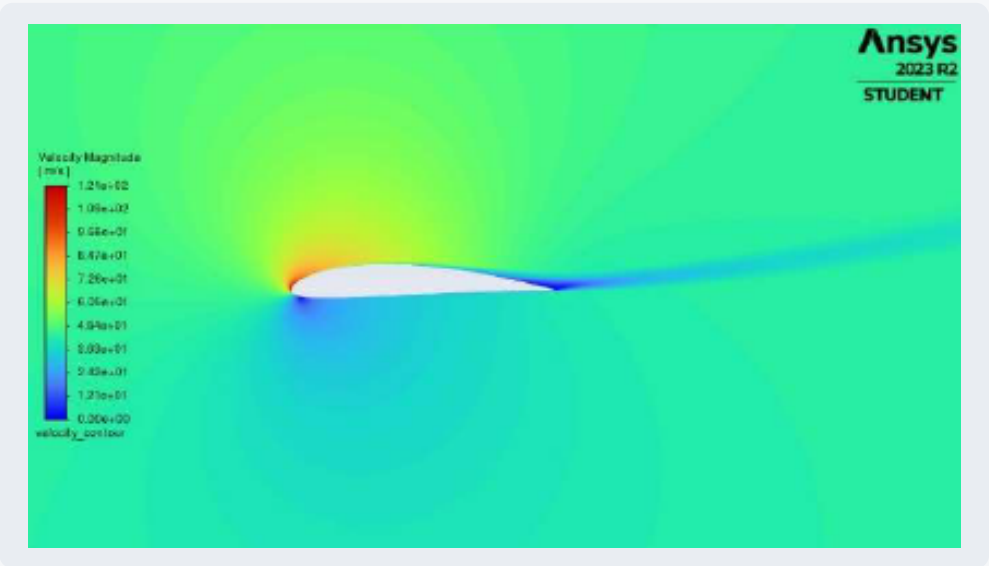
My contribution

- Built ANSYS simulations for convection heat-transfer coefficients and pressure drops.
- Compared ten configurations with varied Reynolds numbers and fin geometry.
- Developed MATLAB ANN models and explored CNN-based geometry recognition.

R > 0.9 describes model correlation, rather than physical product validation.

NACA 4412 aerodynamic analysis

ANSYS Fluent · Spalart–Allmaras turbulence model · February–April 2024



Velocity contour from the airfoil simulation.

At 13°	CFD	Reference
Lift coefficient	1.694	1.721
Drag coefficient	0.0278	0.0286

Engineering question

How closely can a CFD model reproduce reference lift and drag, and how does the flow change as angle of attack increases?

My contribution

- Studied aerodynamic trends from 0° to 13° angle of attack.
- Analyzed pressure distribution, boundary-layer behavior, and flow separation.
- Compared simulated coefficients with NASA reference data.

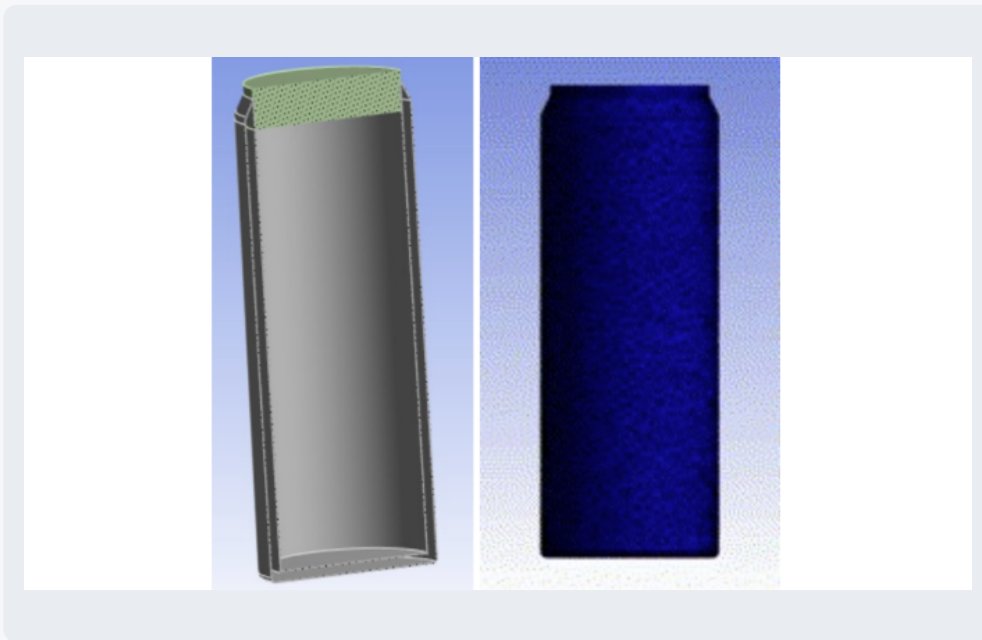
ENGINEERING TAKEAWAY

Reference-data comparisons and flow-field analysis were used together to evaluate the model and interpret aerodynamic trends.

07 / THERMAL RELIABILITY & STATISTICAL MODELING

Vacuum bottle thermal reliability

MATLAB + ANSYS · Material variation, wall thickness, and ambient-temperature effects



Bottle geometry and analysis image.

40

Stainless steel types studied

1M+

Monte Carlo iterations

Engineering question

How do material properties, manufacturing variation, and the surrounding temperature affect the heat lost through a vacuum-insulated bottle?

My contribution

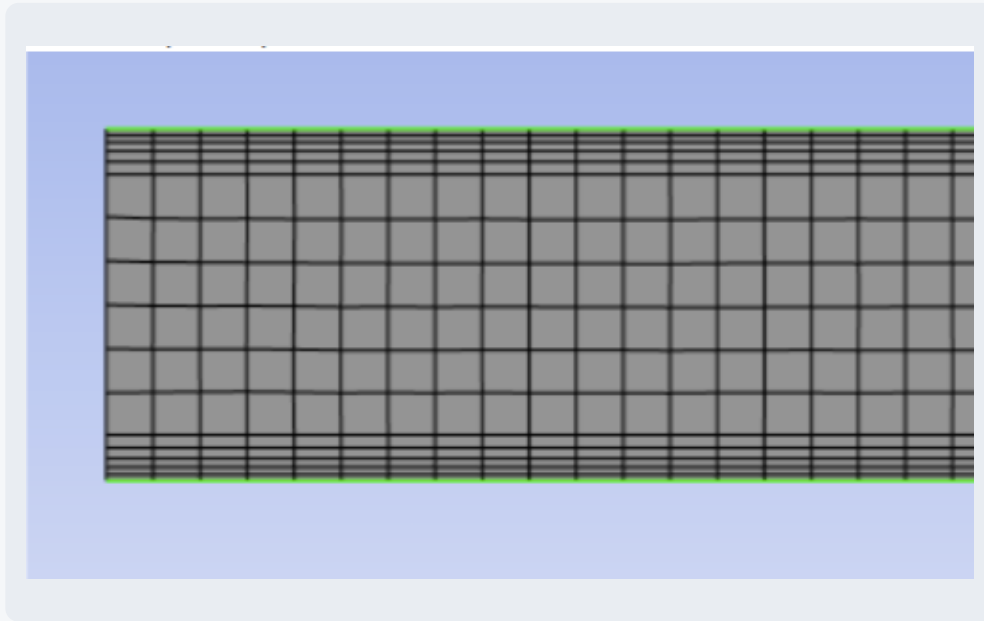
- Studied variation in material thermal conductivity across 40 stainless steel types.
- Used MATLAB and ANSYS to analyze how wall thickness and ambient temperature change heat transfer.
- Ran more than one million Monte Carlo iterations to evaluate thermal-performance variability.

In the winter scenario, 12.03% of simulated cases exceeded the 3.5 W heat-transfer limit, linking insulation reliability to design and ambient conditions.

08 / CFD MODEL VERIFICATION

CFD mesh convergence study

2D convection problem · Grid independence and flow-model comparison



Computational mesh used in the convection study.

5

Mesh element sizes evaluated

3

Flow models compared

Engineering question

How fine does the mesh need to be before the outlet temperature profile stops changing meaningfully?

My contribution

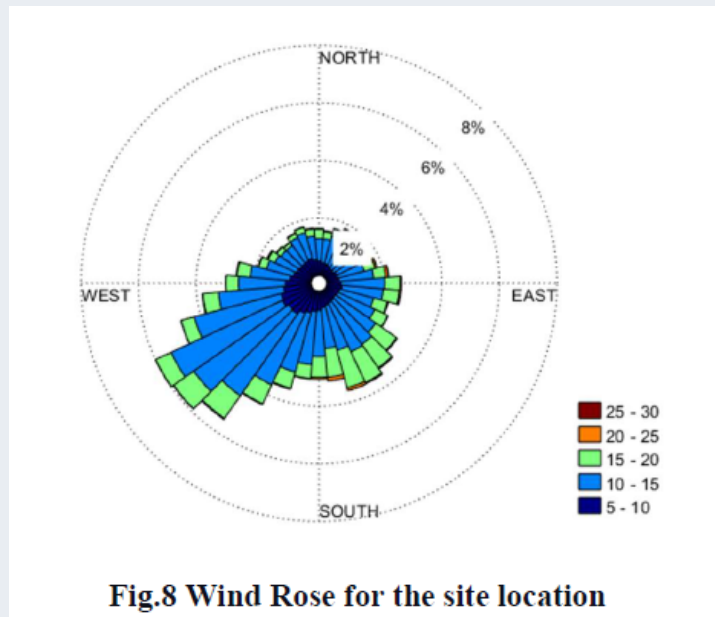
- Compared five mesh element sizes in a 2D convection simulation.
- Extracted temperature profiles along a vertical line at the channel outlet.
- Compared laminar, standard $k-\omega$, and $k-\epsilon$ models using temperature contours and line profiles.

Purpose

Use mesh convergence to support consistent numerical results while avoiding unnecessary computational cost.

Wind farm feasibility & design

Academic design study · Site analysis, turbine selection, and energy-production estimates



Wind rose used to assess the site's wind resource.

3.5 MW

Selected turbine rating

130

Turbines in the proposed layout

The design study

Evaluated a wind farm concept using site wind data, environmental conditions, turbine comparisons, and a proposed layout.

My contribution

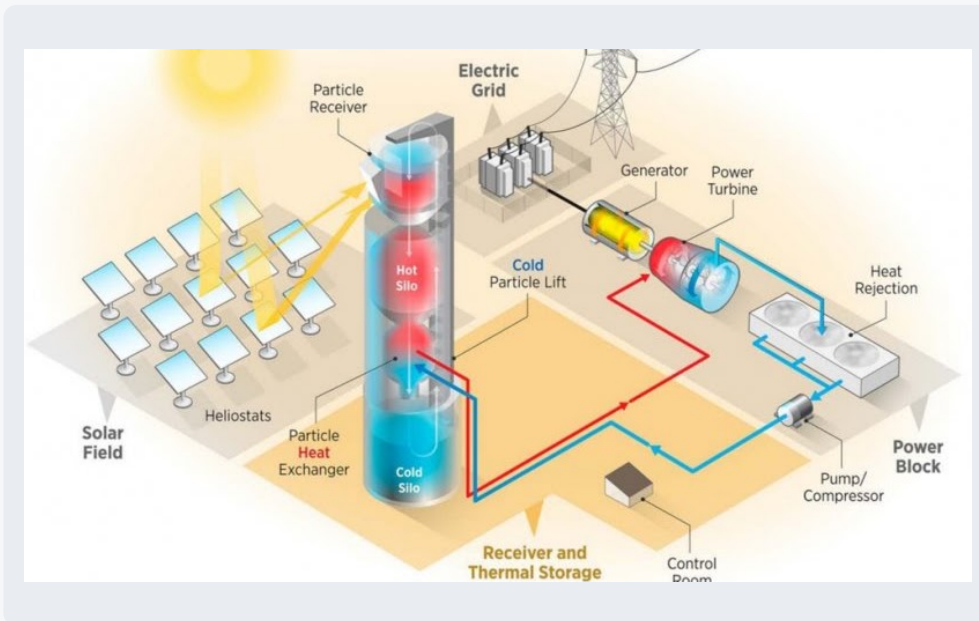
- Used Weibull statistics and wind rose diagrams to analyze wind conditions.
- Compared turbine models and selected a 3.5 MW turbine for the design study.
- Developed a proposed layout with 130 turbines and estimated annual energy production.

The work produced a feasibility assessment and proposed design for renewable energy generation.

10 / CONCENTRATED SOLAR POWER RESEARCH

CSP heat-transfer fluid research

Thermal-management study · Gold nanoplate-infused Dowtherm A



CSP system reference.

RESEARCH FOCUS

Heat-transfer fluid properties and solar thermal performance.

Engineering question

Could adding gold nanoplates to Dowtherm A improve heat-transfer performance in a concentrated solar power system?

My contribution

- Investigated heat-transfer fluids used in concentrated solar power systems.
- Proposed gold nanoplate-infused Dowtherm A as a fluid formulation for comparison.
- Compared the proposed formulation with conventional heat-transfer fluids to assess its thermal-management potential.

The project explored a thermal-management approach to improving CSP system performance.

Double-axis pneumatic welding machine

Academic hardware project · Mechanical design, assembly, and pneumatic actuation



ASSEMBLED PROTOTYPE

Ideal force at 40 kPa (0.4 bar)

Cylinder	Extend	Retract
Rotary	50.3 N	42.2 N
Arm lifting	32.2 N	27.6 N

The project

Engineered and assembled a double-axis machine using rotary and arm-lifting cylinders to explore small-scale welding automation.

My contribution

- Integrated rotary and lifting motion into a physical mechanism.
- Evaluated extending and retracting forces for cylinder selection and actuation.
- Completed the prototype within available resources.

Calculated using 40/16 mm rotary and 32/12 mm lift-cylinder bore/rod diameters. Friction and pressure losses are excluded.

BACKGROUND & CONTACT

Experience summary

Product development · Mechanical design · Manufacturing · Engineering analysis

Experience

Caterpillar

Manufacturing Engineer · Midsys contract

Oct 2024–Oct 2026

Foundry design, commonality, virtual manufacturing, PPAP, MBOM, and Teamcenter.

Arizona State University

Engineering Intern

Jun–Oct 2024

Product layouts, prototyping, six-engineer team leadership, suppliers, and BOM.

Infinite Solutions

Mechanical Design Engineer & Intern

Mar 2020–Jul 2022

Automotive projects, FEA/CFD, thermal and durability analysis, and process workflows.

Tools & capabilities

CAD

SolidWorks, Creo, NX, AutoCAD, Vis Mockup, Rhino, Autodesk Fusion, Autodesk Inventor

Analysis & programming

ANSYS / Fluent, FEA, CFD, Python, MATLAB, SQL, C++

Skills

Product design & prototyping, GD&T, casting, FEA, CFD
BOM / MBOM, PPAP, Teamcenter PLM, design change levels
Virtual manufacturing, design audits, supplier collaboration
Quality assurance, Lean Six Sigma, DMAIC
Process standardization, SOP development

Education

M.S. Mechanical Engineering · Arizona State University, 2024

B.E. Mechanical Engineering · Anna University, 2021

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